Task Categorization and Stressmeter Concept

1. The Idea: A clear and intuitive system to measure the daily workload that takes into account the human psychology and allows the adjustment of the initial condition for accurate estimation and recommendations of the workload distribution.

2. The Situation: Currently, tasks are divided into the following categories and subcategories.

Category	Description	Subcategory
S1	Break	S1: General breaktime, lunch/dinner break S11: Combined breaktime (e.g. cooking, cleaning at breaktime)
S2	Chores	S2: General chores (laundry, ...) S25: S26:
S3	Creative Work	
S4	Girls' Night Out	S4: A pure recovery/going out block S41: A paying job (student job) S42: A volunteering job S45 (discontinued): Fachschaftstreffen S46 (discontinued): Fachschaftsarbeit
S5	Self-Care	S51: Extensive Self Care Sessions
S6	Study	S6: Lectures in Uni, exams S60: Group work in Uni S62: Seminar in Uni
S7	Self-Study Session	S7: General Study Session S75: Extensive Study Session (Exams)
S8	Travel	S8: General Leisure Trip S85: Long Trip (>1000km)
S9	Group Work (unreviewed)	S9: General Group Work (rare) S95 (new): Student organization / Club work

These categories meet the requirements for the integration of "Energy Point", the currency for workload measurement. In its simplest form, an amount of Energy Point is assigned at the start of the day, and the tasks have an influence on this point. The remaining points afterwards will be used to determine the stress level of the day.

3. The Math

Energy is the currency. The maximum energy a day can have is 1500. The default is 1000. Energy is not resetted daily, instead it is transferred from the previous day to the next day.

The base point is the number of points that are always added/subtracted from a task regardless of its duration. This accommodates special types of tasks with very short/long durations with non-linear stress development to time, or tasks with other special mental restraints. For most normal tasks, this point is 0.

The base load factor is the default multiplier for the effect of task duration. The duration of a task is in hours.

The following formula describes the simplest method of calculating the point influence of a single task:

$$Load = basePoint + duration * loadFactor$$

Sign convention: Energy level is positive (only negative for very burned out), a task reduces the points when its energy consuming and adds to the points if it recovers energy. Note: Load is subtracted from the total energy, which means for the following table, negative points mean recovery.

The table below gives the base point and load factor of every subcategory.

Category	Base Point	Load Factor	Load for 60min
S1	0	-100	-100
S11	0	-40	-40
S2	10	60	70
S3	-100	110	10
S4	-100	-75	-175
S41	0	120	120
S42	0	120	120
S5	-145	90	-55
S51	-200	90	-110
S6	0	100	100
S60	0	100	100
S62	0	100	100
S7	0	130	130
S75	20	160	180
S8	-100	-50	-150
S85	-150	-100	-250
S9	30	100	130
S95	60	70	130

4. The energy formula

The energy point at any given time is:

$$E(t) = E(t-1) + \Delta E$$

This function is, however, not distinctive, aka it is not dependent on time alone. In practice, E(0) will be assumed at 1000, and E(t+1) will be calculated with a time step of 1 minute based on the value of E(t). In other words, the energy is updated using the value of the last calculation.

The energy depletion rate at any given time is calculated with the following differential equation:

$$\frac{dE}{dt} = -loadFactor \cdot M_{circadian}(t) \cdot M_{fatigue}(E) \cdot M_{context}(t)$$

5. Extra considerations for more accurate estimation of stress level

5.1. The circadian rhythm multiplier $M_{circadian}$

Concept: Energy cost is dependent on the time in which the task takes place. The function for attention span by time with 2 peaks/24 hours can be simplified to a sinus equation:

$$C(t) = \sin\left(\frac{4\pi \cdot (t - phaseShift)}{24}\right)$$

In which phaseShift is the customizable variable used to align the peaks to match the user's biological clock.

The circadian multiplier: $M_{circadian}(t) = 1 - \beta . C(t)$

C(t) output from -1 (worst performance) to +1 (best performance)
Beta is the strength multiplier. For example at beta = 0.2 and at peak time, then M = 1 – 0.2*(1) = 0.8, which gives a 20% energy reduction.
During circadian peak, tasks feel easier, multiplier goes below 1, during circadian low, tasks feel harder, multiplier goes above 1.

5.2. The fatigue multiplier

Concept: Energy drain scales with actual energy level. The less energy left, the more expensive tasks are.

The formula: $M_{fatigue}(E) = 1 + \alpha \cdot \left(\frac{E_{max}}{E - offset}\right)^{\gamma}$

Emax: 1500

Alpha: 0.1, adjustable

Gamma: 10, adjustable

Offset: -970, adjustable

5.3. Context Multiplier: Not implemented at this development stage. Value is always 1.

6. Recovery mechanism

Energy is always passively (but very slowly) recovered as time passes. The passive recovery is a linear function of time:

$$\Delta E_{passive} = \varepsilon \cdot t$$

In which ε is the passive recovery rate. Defaults to 35 per hour. Adjustable.

Another way (aka the most important way to recover energy) is through sleeping (or idle time). The empty time slot between the end of a task and the beginning of the next task is the idle time. If this time slot is longer than 1 hour, the fast recovery process can take place. The rate of recovery is a differential function:

$$\frac{dE_{fast}}{dt} = k(E_{maxWithCa} - E)$$

k is the idle recovery rate. It has the value of 0.3. Also adjustable.

7. Overload effect: The function above for the recovery rate uses Emaxwithcap. This is the restriction to the most energy that can be recovered in one sleep. The maximum energy to be recovered is:

$$E_{maxWithCa} = E_{base}(1 - \omega \cdot O)$$

In which ω is the penalty strength, defaults to 0.3, adjustable, O ist the overload amount, calculated by the formula:

$$O = \max\left(0, \frac{E_{overload} - E_{end}}{E_{overload}}\right)$$

$E_{overload}$ is the customizable variable that determines from which point the penalty applies. Defaults to 150.

To visualize the above formula, we shall think of sleep as a non-linear function where the energy increases with time, fast at the beginning, slower and slower as the sleep time goes higher, calibrated so that after around 9 hours of sleep the recovery effect is almost negligible. The overload constraint makes sure if you had a really bad day where you extremely overworked, the maximum amount of energy gained during sleep will stop at a lower amount no matter how much you slept.

8. Guideline for designing

The “Stressmeter” Dashboard can be accessed through a badge on the top bar next to the journal island. The badge shows the color-coded value for the immediate Energy point which is updated every minute. Since the value of the energy point for all 1440 steps won’t change until data change, it is advised to save the value and retrieve them every minute instead of recalculating everything. The energy goes from green to yellow to red from high to low. At red, a glowing red border appears on top of the screen. Clicking on the badge triggers a slide down animation revealing the dashboard between the top bar and the main content panel. The main content panel will be pushed down.

The centerpiece of the dashboard is a graph presenting the Energy to time curve. The x-axis is the time from 0 to 24 hours. Behind the curve, render subtle background bands representing tasks. Each task block spans its time interval and is color-coded by category (S1–S9, etc.). These colors should be muted enough not to overpower the main curve but still distinguishable. The alignment between task bands and the energy curve must be precise, reinforcing causality.

A vertical “current time” indicator should be present:

- Positioned at the current time on the x-axis

The graph must support hover interaction:

- When hovering over any point, show a tooltip anchored to the curve
- The tooltip should include:
    - Exact time
    - Energy value
    - Current task/category
    - Current rate of change (e.g., “−2.9 energy/min”)
    - A breakdown of contributing factors:
        - Base load
        - Fatigue multiplier
        - Circadian multiplier
        - Context effects
- The purpose of the tooltip is diagnostic: it should explain *why* the energy is changing, not just show the value

The horizontal guiding line that marks the beginning of the overload phase should also be emphasized.

If data is changed, the app should also recalculate and the graph shall also update in real time.

9. Code requirements:

readable, well-structured code
- strong separation of concerns
- helper functions for each multiplier
- comments explaining the model
- no framework dependency
- pure functions where possible
- robust time parsing helpers
- easy to later integrate into a frontend chart
- avoid overengineering, but do keep it extensible.

10. Numerical details

Use fixed time-step simulation.

Suggested process for each step:
1. determine active task at current time (the function for header train can help)
2. compute base load
3. compute circadian multiplier
4. compute fatigue multiplier using current energy
5. compute context multiplier
6. compute dE/dt
7. integrate one time step forwar
8. clamp energy
9. store point

Be careful with units:
- if base load is defined “per hour”, then dt must be converted to hours
- keep all formulas dimensionally consistent

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11. VERY IMPORTANT DESIGN CONSTRAINTS
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- All effects must work through slope / derivative
- The engine must be interpretable and debuggable
- The implementation should reflect the mathematical model faithfully
- Keep the code modular enough that I can tune coefficients later without rewriting the engine
- Group all adjustable variable to a place where I can conveniently fine tune. Describe their function and effect clearly.

Also include short inline explanations for the formulas you choose.